

Treat the following exercises:

- I. The source file of the following table is an XML document. Rewrite the corresponding XML document source in this condition :
- The root element name is Articles.
 - The node tags' name is the rows' name of the given table.
 - The sub tags' name (of the nodes) is the columns' name of the given table.
 - The sequence sub tag does not exist in the XML document source; it indicates the initial tag's order in this document.

(5 pts)

Articles

	Title	Type	Date	Sequence	notes
CD	My Story	Drama	4-5-1960	2	Film
CD	LOVE's	Romance	7-1-2000	4	Story
Book	Cold War	History	3-8-1990	1	History
Book	OS	System	1-9-1999	5	Scientific
Newspaper	Lsafir	News	1-1-1975	6	Good
Newspaper	Lnahar	News	1-1-1977	3	Good

- II. Write an external DTD file corresponding to the following XML document. In addition, you must define in your DTD file the general entities to reduce the redundancy of tags in the XML document. Then, rewrite the following XML document using in it the entity defined, add the necessary tag to link your new XML document to your DTD file and add two new students.

(6 pts)

```
<?xml version = "1.0" ?>
<University>
  <student>
    <IDS> 101</IDS>
    <Name>Samir Hala</Name>
    <Email>samirhala@gmail.com</Email>
    <Phone>961 03 678 876</Phone>
    <university_Name>Lebanon University</university_Name>
    <Address> Hadat </Address>
    <branch> 1</branch>
  </student>
  <student>
    <IDS> 102</IDS>
    <Name>Salem Diala</Name>
    <Email>dialasalem@gmail.com</Email>
    <Phone>961 71 987 123 </Phone>
    <university_Name>Lebanon University</university_Name>
    <Address> Hadat </Address>
    <branch> 1</branch>
  </student>
  <student>
    <IDS> 103</IDS>
    <Name>Elham Alrais</Name>
    <Email>elhamrais@hotmail.com</Email>
    <Phone>961 70 918 273</Phone>
    <university_Name>Lebanon University</university_Name>
    <Address> Hadat </Address>
    <branch> 1</branch>
  </student>
  ....
</University>
```

- III. Write an XSL file to transform the content of the following XML document into two HTML tables: The first for the French language and the second for the english language. Do not forget to name the table's column like the name of tags. (5 pts)

```
<?xml version= "1.0" encoding= "UTF-16"?>
<schedules>
  <course lang= "FR" code= "X1" >
    <titre> XML </titre>
    <jour> lundi </jour>
    <professeur > Nabil Baze</professeur>
    <temps> 2 H @ 10 AM</temps>
    <sall> B101< /sall>
  </course>

  <course lang= "En" code= "UML1" >
    <title> UML </title>
    <jour> Tuesday </day>
    <teacher > Jamal Azez</teacher>
    <time> 2 H @ 10 AM</time>
    <hall>A102</hall>
  </course>
  <course ...> </course>
  <course ...> </course>
</schedules>
```

- IV. Write the schema file corresponding to the following XML file. In the given XML file, add the necessary name space and link to your schema file into the root element tag in the XML document (Only rewrite the root element tag). (4 pts)

```
<?xml version= "1.0" ?>
<person_address>
  <address type= "Lebanon-address ">
    <name> Samir Alhoss </name>
    <street> Alhamra </street>
    <city> Beirut </city>
    < building > Alariss</building>
    <phone> 01765495</phone>
  </address>
  <address type= "Lebanon-address ">
    <name> Salem Alkodarji </name>
    <street> haret hrake </street>
    <city> Hara </city>
    < building > Alkasfe</building>
    <phone> 01566899</phone>
  </address>
  <address type= "Lebanon-address ">
    <name> Jamila Aldibce </name>
    <street> Aine alromana </street>
    <city> Alay </city>
    <building> Aldibec</building>
    <phone> 05575757</phone>
  </address>
</person_address>
```